

— What Problem Are we Solving?

① $G_{\text{full}} \rightarrow$ Full gaussian primes graph over $R \in \mathbb{Z}$
 R -region of $\mathbb{Z}$

$\{I\} \rightarrow$ set of gaussian primes, $\{0\}$ - also set

As per the next problem we want to prove that
for given R and $k \rightarrow$ no path from $\{I\}$ to $\{0\}$

$\{I\} = \{V_{a_1}, V_{a_2}, \dots, V_{a_n}\} : \forall i=1 \dots n \ V_{a_i} \in G_{\text{full}}$

$\{0\} = \{V_{b_1}, V_{b_2}, \dots, V_{b_m}\} : \forall i=1 \dots m \ V_{b_i} \in G_{\text{full}}$

~~$G_{\text{full}} = \{G_1, G_2, \dots, G_k\}$ - collection of subgraphs~~

" $\sim$ " $\rightarrow$ is equivalence Relation

$u \sim v \rightarrow$ path exists (we can expand it later)

$UF(G) = \{G_1, G_2, \dots, G_k\} : \forall u \in G_k \ \forall v \in G_k \ u \sim v$
and $\forall a, b : a \neq b \ \forall u \in G_a \ \forall v \in G_b \ u \not\sim v$

groups

$UF(G) \rightarrow$ operator, closure

Since $UF_{or}(G) = \{G_1, G_2, \dots, G_k\}$

" $\sim$ " Relation Expansion for sets of vertices

$\{I\} \sim \{O\}$ - means $\exists u \exists v : u \in I \ v \in O : u \sim v$

$\{I\} \not\sim \{O\}$ - means $\forall u \in I \ \forall v \in O \rightarrow u \not\sim v$

Restating in "groups" Notation $\left(\{v\} \in G_k \rightarrow \forall v \in \{v\} \ v \in G_k \right)$

~~$\{I\} \sim \{O\}$ - means $\exists G_k : \forall u \in I \ \forall v \in O \rightarrow u \in G_k \ v \in G_k$~~

~~$\{I\} \not\sim \{O\}$ - means $\exists G_a \exists G_b : \{I\} \in G_a \ \{O\} \in G_b$~~

by the groups definition $\rightarrow$ belonging to different groups
guarantees connectivity absence

Though above there is a contradiction

$\{I\} \sim \{O\} \rightarrow$ means at least one path exists

$\{I\} \not\sim \{O\}$ - No paths exist!

$$UF(G) = \{G_1, G_2, \dots, G_k\}$$

$$\{I\} \sim \{O\} \hookrightarrow \exists u \in \{I\} \exists v \in \{O\} : u \sim v$$

$$\{I\} \sim \{O\} \hookrightarrow \forall u \in \{I\} \forall v \in \{O\} \hookrightarrow u \sim v \quad (1)$$

Restating in "Group Notation" could be several connectivity groups

$$\textcircled{1} \text{ Note that } UF(\{I\}) = \{G_1^I, G_2^I, \dots, G_k^I\} : k = 1 \dots k$$

$$UF(\{O\}) = \{G_1^O, G_2^O, \dots, G_p^O\} : p = 1 \dots p$$

$$\{I\} \sim \{O\} \hookrightarrow \exists u \exists v \exists k : u \in \{I\} v \in \{O\} G_k \in UF(G) : \overset{u \in G_k}{v \in G_k}$$

$$\{I\} \sim \{O\} \hookrightarrow \forall u \in \{I\} \forall v \in \{O\} \nexists k : u \in G_k v \in G_k$$

Introducing $ufs = UF$ search $ufs(v) = k \iff v \in G_k$

$$\{I\} \sim \{O\} \hookrightarrow \forall u \in \{I\} \forall v \in \{O\} ufs(u) \neq ufs(v) \quad (2)$$

$$\nexists k : k \equiv ufs(u) = ufs(v)$$

means that if gaussian Moat exists

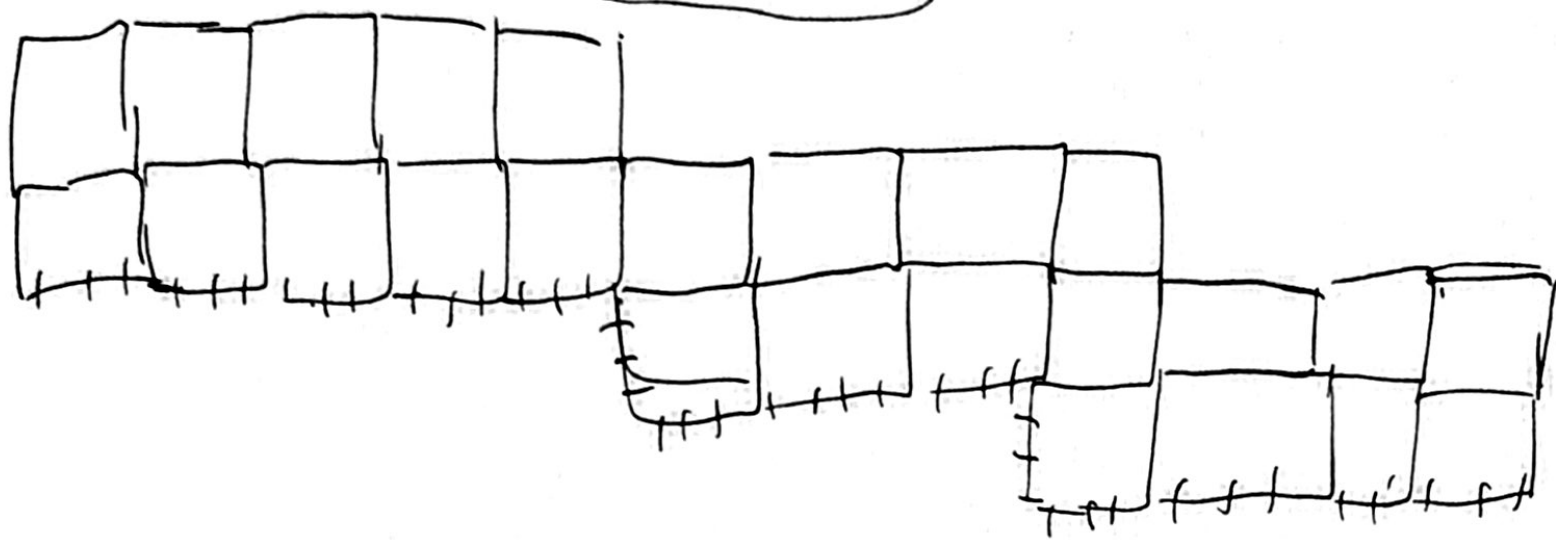
$$\rightarrow UF \text{ founds it} \quad (1) \xleftrightarrow{?} (2)$$

We can prove it both sides

No need Now

$I \rightarrow$ set of inner primes ; O - set of outer primes

Assume $f(I) \rightarrow \{F_1^I, F_2^I, \dots, F_a^I\} \rightarrow$ subgraphs - could overlap
 $: \forall v \in I \exists F_a^I : v \in F_a^I$
 $f(O) \rightarrow \{F_1^O, F_2^O, \dots, F_b^O\}$
 $\forall e = (u, v) \exists F_a^I : e \in F_a^I$
General Partition



$f(I)$ and $f(O) \rightarrow$ "sets of lumps" $\rightarrow$ Generally defined Full partition
 $\{I\} \vee \{O\} \leftrightarrow f(I) \vee f(O)$ (over G)

"Absence of connectivity on Generally defined Full partition is equivalent to absence of path between any two elements of sets"

$f(I) \rightarrow$ set of faces over I $f(O) \rightarrow$ set of faces over O

due to ufs being closure operator $\rightarrow$ ufs over ANY

Generally defined Full partition (the one not losing edges and vertices) is equivalent $ufs(\{I\})$ and $ufs(\{O\})$

UF works for partitions Not loosing edges

assume $P(I) = \{P_1^I, P_2^I, \dots, P_a^I\} : \forall u, v \in I \exists F_a^I : v \in F_a^I$
 $\forall e = (u, v) \in I \exists F_a^I : e \in F_a^I$

$$P(\emptyset) = \{P_1^0, P_2^0, \dots, P_b^0\} \quad \text{---||---||---||---}$$

Its actually simpler

$P_{ort} \neq \text{lump}$ $P_{ort} \rightarrow$ guaranteed to be connectivity group

$P_{ort} \rightarrow$ guaranteed to be single connectivity group (we don't loose)

$P(I) = \{P_1^I, P_2^I, \dots, P_a^I\} : 1) \forall v \in I \exists F_a : v \in F_a \text{ (one } \neq V \text{)}$
 $2) \forall u, v \in I \rightarrow u \sim v$

We loose SOME edges with ports

if for $\forall i = 1 \dots a \forall j = 1 \dots b \quad \underline{ufsg(P_i^I)} \neq \underline{ufsg(P_j^0)}$

$\downarrow$ since $\forall u, v \in P \quad u \sim v \rightarrow$

$$\rightarrow \boxed{ufsg(u) = ufsg(v) = ufsg(P)}$$

$$\downarrow \boxed{I \sim \emptyset}$$

$$P(I) = \{P_1^I, P_2^I, \dots, P_a^I\} \quad | \quad P(O) = \{P_1^O, P_2^O, \dots, P_b^O\}$$

- 11-11 -

- 1) no gaussian primes lost
- 2) Single port is guaranteed to be one connectivity group
- 3*) one gaussian prime could be in more than 1 port

$UFS_G \rightarrow$ global UF groups assignment

$$\forall u, v \in P_a^I \hookrightarrow UFS_G(u) = UFS_G(v) \stackrel{\text{definition}}{=} UFS_G(P_a^I)$$

by UFS_G definition for ports UFS verdict that Ports are Not connected is equivalent to $\{O\} \not\sim \{I\}$

Port creation could be done by Applying UF over the tile area